

human

Show me all the buildings that are within 1km of (291560 , 092966) given as easting and northing.

host_agent

Tool Calls

- Calling tool `generate_metadata_for_all_artifacts` with args: `{}`
- Tool `generate_metadata_for_all_artifacts` returned result: `{}`
- Calling tool `send_message` with args: `{ 'target': 'named_area', 'task_description': 'Find the named area or polygon that contains the point (291560, 092966) in easting and northing coordinates.' }`
- Tool `send_message` returned result: Message sent to agent `named_area`

named_area_agent

Tool Calls

- Calling tool `call_os_ngd` with args: `{ 'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': False, 'filename': 'named_area_search_result' }`
- Tool `call_os_ngd` returned result: Search results have been found. Artifact generated is: `named_area_search_result`, Description: A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the named area or polygon that contains the point (291560, 092966) in easting and northing coordinates. Count of records fetched: 98.
- Calling tool `send_message` with args: `{ 'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'named_area_search_result' to find the polygon that contains the point (291560, 092966). Save the filtered results as 'filtered_named_area_result'." }`
- Tool `send_message` returned result: Message sent to agent `data_analysis_agent`

data_analysis_agent

Tool Calls

- Calling tool `generate_metadata_for_artifacts` with args: { 'artifact_names': ['named_area_search_result'] }
- Tool `generate_metadata_for_artifacts` returned result: Metadata generated for 'named_area_search_result'.

Tool Code Executor

```
import geopandas as gpd
from shapely.geometry import Point

def filter_polygon_containing_point(data: list):
    gdf = data[0]
    # Create the point with the given coordinates
    point = Point(291560, 92966)
    # Filter polygons that contain the point
    filtered = gdf[gdf['geometry'].contains(point)]
    summary = f"Filtered polygons containing the point (291560, 92966). Number of matching polygons: {len(filtered)}."
    artifact_name = 'filtered_named_area_result'
    artifact_description = 'GeoDataFrame containing polygons from named_area_search_result that contain the point (291560, 92966).'
    return [summary, artifact_name, artifact_description, filtered]
```

(First three attempts failed due to variable name errors, only the last version worked.)

Tool `code_executor` returned result: ['Filtered polygons containing the point (291560, 92966). Number of matching polygons: 1.',]

Tool `send_message` returned result: Filtered polygons containing the point (291560, 92966). Number of matching polygons: 1.

An artifact has been generated with the name: `filtered_named_area_result` and the description: `GeoDataFrame containing polygons from named_area_search_result that contain the point (291560, 92966).`

named_area_agent

The search has been completed, and the polygon containing the point (291560, 092966) has been identified. The filtered results are saved in the artifact named `filtered_named_area_result`.

Tool Calls

- Calling tool `send_message` with args: { 'target': 'land_features', 'task_description': 'Find the land feature polygon or area that contains the point (291560, 092966) in easting and northing coordinates.' }
- Tool `send_message` returned result: Message sent to agent `land_features`

land_features_agent

I cannot solve this query as it does not mention a specific area or bounding box to search within. Please provide the name of the area or bounding box to proceed.

host_agent

Tool Calls

- Calling tool `send_message` with args: { `'target': 'buildings', 'task_description': "Find all buildings within the area defined by the artifact 'filtered_named_area_result'."` }
- Tool `send_message` returned result: Message sent to agent buildings

buildings_agent

Tool Calls

- Calling tool `call_os_ngd` with args: { `'filters': None, 'bbox': 'filtered_named_area_result', 'polygon_or_point': True, 'street_address': False, 'filename': 'all_buildings_in_filtered_area'` }
- Tool `call_os_ngd` returned result: Multiple search results have been found in multiple datasets. Artifacts generated are: ['buildings_all_buildings_in_filtered_area', 'buildingline_all_buildings_in_filtered_area', 'buildingpart_all_buildings_in_filtered_area'], Descriptions: ["A geopandas dataframe containing building data with filters and bbox applied for the query Find all buildings within the area defined by the artifact 'filtered_named_area_result'. using filters None.", ...], counts of records fetched: [69169, 69342, 85480].

I found the following artifacts for buildings within the area defined by 'filtered_named_area_result':

1. **buildings_all_buildings_in_filtered_area**: A geopandas dataframe containing building data.
2. **buildingline_all_buildings_in_filtered_area**: A geopandas dataframe containing building line data.
3. **buildingpart_all_buildings_in_filtered_area**: A geopandas dataframe containing building part data.

Would you like any further analysis or filtering on these results?

host_agent

Tool Calls

- Calling tool `send_message` with args: { 'target': 'plotting_agent', 'task_description': "Plot all buildings from 'buildings_all_buildings_in_filtered_area' that are within 1km of the point (291560, 092966) in easting and northing coordinates." }
- Tool `send_message` returned result: Message sent to agent `plotting_agent`

plotting_agent

Tool Calls

- Calling tool `generate_metadata_for_artifacts` with args: { 'artifact_names': ['buildings_all_buildings_in_filtered_area'] }
- Tool `generate_metadata_for_artifacts` returned result: Metadata generated for 'buildings_all_buildings_in_filtered_area'.

Tool Code Executor

```
def plot_buildings_within_1km_of_point(data: list):
    import geopandas as gpd
    from shapely.geometry import Point
    import folium
    # Assume data[0] is the buildings GeoDataFrame
    gdf = data[0]
    # The point in easting/northing (assume EPSG:27700 British National Grid)
    point = Point(291560, 92966)
    # Ensure CRS is set
    if gdf.crs is None or gdf.crs.to_epsg() != 27700:
        gdf = gdf.set_crs(epsg=27700, allow_override=True)
    # Create 1km buffer around the point
    buffer = gpd.GeoSeries([point], crs='EPSG:27700').buffer(1000)
    # Select buildings within 1km
    buildings_within_1km = gdf[gdf.intersects(buffer.iloc[0])].copy()
    # Convert to WGS84 for folium
    buildings_within_1km = buildings_within_1km.to_crs(epsg=4326)
    buffer_wgs84 = buffer.to_crs(epsg=4326)
    point_wgs84 = gpd.GeoSeries([point], crs='EPSG:27700').to_crs(epsg=4326)
    # Remove non-serializable columns (like Timestamps)
    drop_cols = [col for col in buildings_within_1km.columns if 'date' in col or 'evidencedate' in col or 'updatedate' in col]
    buildings_within_1km = buildings_within_1km.drop(columns=drop_cols)
    # Create folium map centered at the point
    center = [point_wgs84.geometry.iloc[0].y, point_wgs84.geometry.iloc[0].x]
    m = folium.Map(location=center, zoom_start=15)
```

```
# Plot buffer
folium.GeoJson(buffer_wgs84.geometry.iloc[0], name='1km Buffer', style_function=lambda x: {'color': 'blue', 'fill': False, 'weight': 2}).add_to(m)
# Plot point
folium.Marker(location=center, popup='Query Point', icon=folium.Icon(color='red')).add_to(m)
# Plot buildings
folium.GeoJson(buildings_within_1km, name='Buildings', style_function=lambda x: {'color': 'green', 'fill': True, 'weight': 1, 'fillOpacity': 0.3}).add_to(m)
folium.LayerControl().add_to(m)
# Save map
map_filename = 'buildings_within_1km_of_point.html'
m.save(map_filename)
# Prepare summary
summary = f"Found {len(buildings_within_1km)} buildings within 1km of the point (291560, 92966). First 5 entries:\n" + str(buildings_within_1km[['osid', 'description', 'buildinguse', 'geometry_area_m2'][:5]])
return [summary, map_filename, 'Folium map showing all buildings within 1km of the specified point, with buffer and point marked.', map_filename]
```

- Tool code_executor returned result: ['Found 6241 buildings within 1km of the point (291560, 92966). First 5 entries:\n osid description \ 41930 dbf62c8e-7b7d-41c7-9522-367e4f3b99da End-Of-Terrace House 4179 17e2dae8-361b-4af6-8fce-ce3262b098c4 Mid-Terrace House 6314 259c27a0-00eb-41f8-a8cf-a0b319e13b0d Commercial Building 30675 bed4da67-1f00-47ca-bb79-2e1ebd8a8327 Ancillary Building 49727 ee00bed3-e6b2-4721-a177-031fd26a3c4f Commercial Building \n buildinguse geometry_area_m2 \n41930 Residential Accommodation 55.697 \n4179 Residential Accommodation 50.627 \n6314 Commercial Activity: Retail 1502.257 \n30675 Unknown 315.052 \n49727 Commercial Activity: Retail 442.250 ',]

Found 6241 buildings within 1km of the point (291560, 92966). First 5 entries:

	osid	description	buildinguse	geometry_area_m2
41930	dbf62c8e-7b7d-41c7-9522-367e4f3b99da	End-Of-Terrace House	Residential Accommodation	55.697
4179	17e2dae8-361b-4af6-8fce-ce3262b098c4	Mid-Terrace House	Residential Accommodation	50.627
6314	259c27a0-00eb-41f8-a8cf-a0b319e13b0d	Commercial Building	Commercial Activity: Retail	1502.257
30675	bed4da67-1f00-47ca-bb79-2e1ebd8a8327	Ancillary Building	Unknown	315.052
49727	ee00bed3-e6b2-4721-a177-031fd26a3c4f	Commercial Building	Commercial Activity: Retail	442.250

Artifact: buildings_within_1km_of_point.html

Description: Folium map showing all buildings within 1km of the specified point, with buffer and point marked.

Map file: buildings_within_1km_of_point.html

host_agent

All buildings within 1km of the point (291560, 092966) have been identified and plotted. There are 6,241 buildings within this range. The map highlights:

- The specified point (291560, 092966) in easting and northing.
- A 1km buffer around this point.
- All buildings within this buffer.

You can view the results in the artifact: [buildings_within_1km_of_point.html](#), which is an interactive map showing all relevant buildings, the buffer, and the point.

If you need further filtering (e.g., by building type or use), or want to see more details, let me know!